The Geometry of What Maps Forget

Residual kernels, observable duality, and exact resolution complexity

Nova Spivack

Independent Researcher

March 2026

Keywords: residual kernel, information loss, fiber automorphism, observable duality, resolution complexity, graph coloring, Lean 4 formalization.
MSC 2020: 03B70, 03F40, 18A05, 05C15.
Code availability: Lean 4 formalization at `reflexive-architecture-lean`. Build: `lake build`. Zero `sorry` in proof terms. See `MANIFEST.md` for the complete theorem index.

Abstract

Every non-injective function forgets something. We develop a universal theory of *what* it forgets and *how much* it costs to recover. For an arbitrary map $\pi : \mathsf{Realized} \to \mathsf{Bare}$, five conditions—non-exhaustion, non-injectivity, existence of a non-trivial π-preserving endomorphism, witness diversity, and nonemptiness of the residual kernel—are proved equivalent. Either all hold simultaneously, or none do (all-or-nothing dichotomy).

We identify the *residual kernel* $\mathcal{K}(\pi)$ as the canonical object of information loss: the set of distinct domain pairs sharing a common image. We prove its complete geometry: it decomposes fiberwise into complete indistinguishability cliques; bare-determined observables form a Boolean algebra dual to kernel separation; and refinement acts as graph coloring of fiber cliques, with exact per-fiber complexity (resolving a fiber of size n requires exactly n colors). The comparison map is the universal coarse observable; the structure is functorial; and the entire theory holds unconditionally for every function between any two types.

Because the theory is stated for an arbitrary function between arbitrary types, it applies not only to certification but to any map that compresses structure: group homomorphisms, topological quotients, type-theoretic projections, measurements, compilers, and hash functions all inherit the full residual geometry. The theory is a universal geometry of information loss under any map.

All results are formalized in Lean 4 [8] with Mathlib [9], with zero `sorry` in proof terms. The theory is discharged against the Infinity Compression flagship and admits instances from algebra, topology, and computability.

Contents

1 Introduction **2**
1.1 Main theorem package . . . 3
1.2 Novelty by synthesis . . . 4
1.3 Relationship to the earlier summit . . . 4

2 Reflective certification systems **4**

3 The fundamental equivalence theorem **5**

4 The exhaustion criterion **6**

5 The adequacy summit **7**

6 The residual kernel **8**

7 Fiber geometry **9**

8 Observable duality **9**

9 Refinement and resolution **10**

10 Optimal certification and residual measurement **12**

11 Functoriality and invariance **13**

12 The fundamental theorem of residual geometry **14**

13 Source discharges **14**

14 Relationship to incompleteness and proof relevance **15**

15 The universal theory of forgetting **16**

16 Frontier **17**

17 Conclusion **18**

A Theorem index and Lean correspondence **18**

1 Introduction

This paper develops a universal theory of what maps forget.

Every function $\pi : \mathsf{Realized} \to \mathsf{Bare}$ between two types identifies some elements of $\mathsf{Realized}$: those that share an image in Bare. When π is injective, nothing is lost. When it is not, the *residual*—the totality of distinctions that π cannot see—becomes a nontrivial mathematical object.

We prove that the residual is not merely "some leftover." It is a *canonically organized space of indistinguishability* with a precise internal geometry, exact observability laws, a resolution theory governed by graph coloring, and sharp quantitative bounds on the cost of eliminating it.

The theory arose from a certification problem (where Bare represents bare certificates and $\mathsf{Realized}$ represents realized witnesses), but the mathematics depends only on the function π. Group homomorphisms, topological quotients, type-theoretic projections, measurements, compilers, and hash functions all inherit the full geometry. We use the language of certification throughout, but the results are universal.

The results ascend through a sequence of summits:

Summit I. Non-exhaustion exists. Adequate systems—those carrying witness-diversity data—force non-exhaustion as a theorem, not an assumption (§5).

Summit II. The residual has a canonical object. The *residual kernel* $\mathcal{K}(\pi)$ is the set of distinct realization pairs sharing a bare certificate. It is nonempty if and only if the system is non-exhaustive (§6).

Summit III. The residual has geometry. Fibers are the connected components of the kernel graph, and each fiber is a complete indistinguishability clique. Observables and refinements are governed by kernel structure (§§7–8).

Summit IV. Resolution has exact complexity. Refinement acts as graph coloring of fiber cliques. Resolving a fiber of size n requires exactly n colors. Terminal resolution is singletonization of refined classes (§9).

Summit V. The fundamental equivalence. Non-exhaustion, non-injectivity, witness diversity, fiber automorphism, and nontrivial kernel are five equivalent characterizations of the same phenomenon. Either all hold, or none do (§3).

Summit VI. Universality. The entire theory applies to any function between any two types. "Certification" is a motivating interpretation, not a formal hypothesis. The residual geometry is a universal theory of information loss (§15).

Expository order follows the ascent from existence to geometry to resolution; the logical organizing theorem is the Classification Theorem in §3, which governs all five summits simultaneously.
All theorems are formalized in Lean 4 [8] with Mathlib [9] dependencies, building cleanly with zero `sorry`. The formalization currently comprises approximately 235 theorems across 26 modules.

1.1 Main theorem package

The paper establishes the following five named results, each holding unconditionally for every function between any two types. Logically, the Classification Theorem governs the whole theory; expositionally, we ascend through the canonical object, its geometry, and its resolution before stating the equivalence.

(A) **Classification Theorem** (Theorem 3.1): non-exhaustion, non-injectivity, witness diversity, fiber automorphism, and nontrivial kernel are equivalent. Either all hold, or none do.

(B) **Vanishing Criterion** (Theorem 4.1): the system is exhaustive iff the kernel is empty iff every fiber is a subsingleton.

(C) **Residual Geometry Theorem** (Theorem 12.1): the residual kernel decomposes fiberwise into complete cliques; observables are dual to kernel separation; refinement resolves iff it separates every kernel pair; resolution decomposes into independent per-fiber coloring problems.

(D) **Resolution Complexity Theorem** (Theorem 9.5): resolving a fiber of size n requires exactly n distinct refinement values on that fiber.

(E) **Certification–Refinement Gap Theorem** (Theorem 9.16): certification cannot resolve its own residual; refinement can, but at measurable cost; for adequate systems the cost is always positive.

These are not five independent results but a single coherent theory: (A) classifies the phenomenon, (B) characterizes its absence, (C) describes its geometry, (D) quantifies the cost of eliminating it, and (E) proves that the cost is real and irreducible for adequate systems. Because the theory is stated for an arbitrary function between arbitrary types, it applies to any map that compresses structure (§15).

Logical vs. expository order. Logically, the Classification Theorem governs the whole theory; expositionally, we ascend through the canonical object, its geometry, and its resolution before returning to that equivalence as the governing law.

Question	Answer
What is information loss?	Nontrivial residual kernel
When does it occur?	Iff any of five equivalent conditions hold
Where does it live?	In the fibers (complete kernel cliques)
What can see it?	Exactly the non-bare-determined predicates
How is it resolved?	By refinements separating kernel pairs
What does resolution cost?	Exactly fiber-size-many colors
When is nothing left?	Exactly when π is injective
Can the map see its own loss?	No (meta-level invisibility)

1.2 Novelty by synthesis

The novelty of this paper is not the isolated appearance of fibers, kernels, graph colorings, or factorization lemmas—each of those ingredients has antecedents. The novelty is that these ingredients form *one exact, machine-checked theory*: a classification theorem for information loss, a canonical residual object, a duality between observables and kernel separation, an exact resolution criterion, a quantitative complexity law, and source discharges from independent theorem programs—all valid for arbitrary maps, all formalized with zero `sorry`. No previous work has assembled this particular combination into a unified theory, and the assembly is not routine: the Classification Theorem requires a classical construction (fiber swap) that closes the circle between five superficially unrelated conditions; the Residual Geometry Theorem packages unconditional structure that exists for every function; and the formal verification is not decoration but the mechanism by which the synthesis is checked. The central claim is therefore not merely that non-injective maps have fibers, but that arbitrary maps carry a full residual geometry with a classification theorem, a canonical residual object, an observable duality, and an exact resolution complexity law.

1.3 Relationship to the earlier summit

Earlier work in Infinity Compression established the *first summit* of the program: canonical certification does not exhaust reflective realization in a concrete flagship setting. A companion synthesis paper [19] then showed that this phenomenon sits inside a larger three-layer architecture linking NEMS, APS, and Infinity Compression—with bridge theorems, the non-erasure principle, and machine-checked coherence axioms proved definitionally.

The present paper is the *higher summit forced by that earlier work*. It extracts from those source discharges the map-theoretic core that survives after domain-specific structure is stripped away. What appeared first as irreducible residue above canonical certification in a concrete program is shown here to be an instance of a universal geometry of information loss under arbitrary maps. Summit 1 established that non-exhaustion is real and unavoidable; Summit 2 identifies the exact mathematical object it produces, proves its complete geometry, and shows that the phenomenon belongs to all functions, not just to the flagship architecture.

In that sense this paper does not supersede the earlier summit. It completes its abstraction. The old flagship now appears as a source discharge of the universal theory (§13).

2 Reflective certification systems

We work in a universe-polymorphic setting. Fix types Bare and $\mathsf{Realized}$ in a common universe u. For the universal theory, an RCS is simply a convenient packaging of an arbitrary map together with two inert fields retained for source-level compatibility (see Remark 2.2).

Definition 2.1 (Reflective certification system)**.** A *reflective certification system* S on carriers $(\mathsf{Bare}, \mathsf{Realized})$ consists of:

- a *comparison map* $\pi : \mathsf{Realized} \to \mathsf{Bare}$,
- a *canonical region* $\mathrm{Canonical} : \mathsf{Bare} \to \mathrm{Prop}$,
- a *soundness predicate* $\mathrm{Sound} : \mathrm{Prop}$.

Remark 2.2 (What drives the universal theory). Every theorem in §§3–12 depends only on the comparison map $\pi : \mathsf{Realized} \to \mathsf{Bare}$. The additional fields Canonical and Sound are retained because they are part of the originating formal interface and matter for source discharges (§13), but they play no role in the universal results. This is what makes the theory broad: it applies to any function between any two types, regardless of what mathematical structures the carriers carry.

In the universal reading of this paper, one may equivalently regard an RCS as an arbitrary map packaged in the originating interface vocabulary; the name "reflective certification system" is inherited from the source program and is not a restriction on the mathematics. We write π_S or simply π for the comparison map of S.

Definition 2.3 (Exhaustion predicates)**.** Let S be an RCS with comparison map π.
- S is *minimally exhaustive* (or simply *exhaustive*) if π is injective: $\pi(x) = \pi(y) \Rightarrow x = y$.
- S is *strongly exhaustive* if π admits a global section: $\exists\, s : \mathsf{Bare} \to \mathsf{Realized},\ \pi \circ s = \mathrm{id}$.
- S is *non-exhaustive* if there exist $x, y \in \mathsf{Realized}$ with $x \neq y$ and $\pi(x) = \pi(y)$.

Definition 2.4 (Fiber)**.** The *fiber* of π over $b \in \mathsf{Bare}$ is $\mathrm{Fib}(b) = \pi^{-1}(b) = \{\, x \in \mathsf{Realized} \mid \pi(x) = b \,\}$.

Proposition 2.5 (Classical dichotomy)**.** *For any RCS S, exactly one of the following holds: S is exhaustive, or S is non-exhaustive.*

Proof. By classical logic applied to $\mathrm{Injective}(\pi)$. □

Example 2.6 (Toy non-exhaustive system)**.** Let $\mathsf{Bare} = \mathsf{Bool}$, $\mathsf{Realized} = \mathsf{Bool} \times \mathsf{Bool}$, and $\pi = \mathrm{fst}$. Then $\mathrm{Fib}(\mathsf{true}) = \{(\mathsf{true}, \mathsf{false}), (\mathsf{true}, \mathsf{true})\}$ has two elements, so the system is non-exhaustive. The residual kernel contains the pair $((\mathsf{true}, \mathsf{false}), (\mathsf{true}, \mathsf{true}))$. The swap $\tau(a, b) = (a, \neg b)$ is a non-identity π-preserving endomorphism, illustrating the fiber automorphism of Theorem 3.1(iii). Resolving requires a refinement with at least two values on each nontrivial fiber.

Example 2.7 (Identity collapse)**.** Let $\mathsf{Bare} = \mathsf{Realized} = \mathsf{Bool}$ and $\pi = \mathrm{id}$. Every fiber is a singleton, the kernel is empty, and the system is exhaustive. No adequate or reflectively sufficient structure exists on this system.

Remark 2.8 (Lean formalization).
- `ReflectiveCertificationSystem` in `Universal/ReflectiveCertificationSystem.lean`
- `MinimalExhaustive`, `StrongExhaustive`, `NonExhaustive` in `Universal/ExhaustionDefinitions.lean`
- `minimal_exhaustive_or_nonExhaustive_classical` in `Universal/Dichotomy.lean`
- `Fiber`, `IsSection`, `LiftableWith` in `Universal/FiberBasics.lean` and `Universal/SectionsAndObstructions.lean`

3 The fundamental equivalence theorem

The deepest result of this paper is that five superficially different conditions on an RCS are logically equivalent. We state it early because it governs everything that follows.

Theorem 3.1 (Classification Theorem)**.** *For any RCS $S = (\mathsf{Bare}, \mathsf{Realized}, \pi)$, the following are equivalent:*

(i) Non-exhaustion. $\exists\, x, y \in \mathsf{Realized},\ x \neq y \ \wedge\ \pi(x) = \pi(y)$.

(ii) Non-injectivity. $\neg\, \mathrm{Injective}(\pi)$.

(iii) Fiber automorphism. $\exists\, \tau : \mathsf{Realized} \to \mathsf{Realized},\ \pi \circ \tau = \pi \ \wedge\ \tau \neq \mathrm{id}$.

(iv) Witness diversity. $\exists\, w_1, w_2 \in \mathsf{Realized},\ w_1 \neq w_2 \ \wedge\ \pi(w_1) = \pi(w_2)$.

(v) Nontrivial kernel. $\mathcal{K}(\pi) \neq \varnothing$.

Proof. The equivalences (i)⇔(ii), (i)⇔(iv), and (i)⇔(v) are definitional: non-exhaustion *is* witness diversity (the definitions coincide), non-exhaustion is the negation of injectivity by the classical dichotomy, and the kernel is nonempty exactly when a witness pair exists. These equivalences are recorded for completeness; the theorem's content lies in the non-trivial direction.

The substantive direction is (i)⇒(iii). Given $x \neq y$ with $\pi(x) = \pi(y)$, define the *fiber swap*

$$\tau(z) \;=\; \begin{cases} y & \text{if } z = x, \\ x & \text{if } z = y, \\ z & \text{otherwise.} \end{cases}$$

Then $\pi(\tau(z)) = \pi(z)$ for all z (checking the three cases using $\pi(x) = \pi(y)$), and $\tau \neq \mathrm{id}$ because $\tau(x) = y \neq x$. This construction uses classical decidability of equality on $\mathsf{Realized}$ (via `Classical.em` in the Lean formalization). The converse direction (iii)⇒(i) is constructive.

The converse (iii)⇒(i): if $\tau \neq \mathrm{id}$, there exists x with $\tau(x) \neq x$; then $\pi(\tau(x)) = \pi(x)$ with $\tau(x) \neq x$, witnessing non-exhaustion. □

The direction (i)⇒(iii) is the *fiber automorphism diagonal*: the comparison geometry of any non-exhaustive system *forces* the existence of a non-trivial π-preserving self-map. The system's own structure generates the symmetry that prevents exhaustion.

Theorem 3.1 is the *Classification Theorem for information loss under a map.* The equivalence of the five conditions is not obvious *a priori*: non-injectivity is a property of a function, witness diversity is data carried by a structure, fiber automorphism is an endomorphism condition, and kernel nonemptiness is a set-theoretic statement. That these four perspectives collapse into a single dichotomy—and that the fiber automorphism is *forced* by non-exhaustion via a classical construction—is the theorem's content. Non-exhaustion is not merely a yes/no property; it is a condition with five equivalent faces, each revealing a different aspect of the same underlying geometry.

Theorem 3.2 (All-or-nothing dichotomy)**.** *For any RCS S, either all five conditions in Theorem 3.1 hold simultaneously, or none do. There is no intermediate position.*

Remark 3.3 (Honest ceiling). By the equivalence (i)⇔(ii), any predicate P satisfying $P(S) \Rightarrow \mathrm{NonExhaustive}(S)$ also satisfies $P(S) \Rightarrow \neg\,\mathrm{Injective}(\pi_S)$. This is logically immediate, but its *methodological* content is significant: at the level of bare comparison maps, every forcing predicate must imply non-injectivity. There is no way to force non-exhaustion without implying failure of injectivity. So any summit class must carry structural data (witnesses, twists, cardinality gaps) rather than a clever predicate on π alone.

Remark 3.4 (Lean formalization).
- `equiv_1_iff_2` through `equiv_1_iff_5`, `all_or_nothing`, `FiveWayEquivalence`, `fiveWayEquivalence`, `nonExhaustive_implies_fiberAutomorphism` in `Summit/FundamentalEquivalence.lean`
- `forcing_implies_not_injective` in `Summit/Adequacy.lean`

4 The exhaustion criterion

The exhaustion criterion is the exact dual vanishing theorem to the fundamental equivalence: where Theorem 3.1 classifies the *presence* of certification failure, the exhaustion criterion classifies its *absence*.

Theorem 4.1 (Vanishing Criterion)**.** *For any RCS S, the following are equivalent:*

(i) π *is injective.*

(ii) $\mathcal{K}(\pi) = \varnothing$.

(iii) Every fiber $\mathrm{Fib}(b)$ *is a subsingleton.*

(iv) S *is not non-exhaustive.*

There is no partial exhaustion: the system is completely exhaustive if and only if the kernel is empty, if and only if every fiber has at most one element.

Remark 4.2 (Lean formalization).

- `exhaustive_iff_kernel_empty`, `kernel_empty_iff_all_fibers_subsingleton`, `exhaustive_iff_not_nonExhaustive` in `Residual/FundamentalTheorem.lean`

5 The adequacy summit

The fundamental equivalence tells us *what* non-exhaustion is equivalent to. The adequacy summit tells us *when* it is forced.

Definition 5.1 (Adequate reflective system)**.** An *adequate reflective system* is an RCS equipped with *witness diversity data*: two distinguished elements $w_1, w_2 \in \mathsf{Realized}$ with $w_1 \neq w_2$ and $\pi(w_1) = \pi(w_2)$.

This is *data*, not a proposition: the witnesses are fields of the structure, not existentially quantified. The distinction matters because it gives the class computational content.

Theorem 5.2 (Adequacy Theorem)**.** *Every adequate reflective system is non-exhaustive.*

Proof. The two witnesses are the existential witnesses for non-exhaustion. □

Proposition 5.3 (Non-circularity)**.** *The axioms of adequacy mention neither non-exhaustion nor injectivity. Identity-style collapse ($\pi = \mathrm{id}$) provably cannot satisfy the axioms: $\pi(w_1) = \pi(w_2)$ with $\pi = \mathrm{id}$ forces $w_1 = w_2$, contradicting distinctness. More generally, any adequate system has non-injective comparison.*

Definition 5.4 (Reflectively sufficient system)**.** A *reflectively sufficient system* is an RCS equipped with a non-identity π-preserving endomorphism $\tau : \mathsf{Realized} \to \mathsf{Realized}$ satisfying $\pi \circ \tau = \pi$ and $\tau \neq \mathrm{id}$.

Theorem 5.5 (Self-generation)**.** *Every reflectively sufficient system is adequate. Since $\tau \neq \mathrm{id}$, there exists x with $\tau(x) \neq x$; then $(x, \tau(x))$ is a witness pair with $\pi(x) = \pi(\tau(x))$.*

The self-generation theorem shows that witness diversity need not be externally supplied: the system's own internal symmetry generates it. This is the *Gödelian core* of the program. In Gödel's incompleteness theorem [1], a sufficiently rich formal system's own expressive power constructs a sentence the system cannot decide. Here, a non-exhaustive system's own comparison geometry constructs the endomorphism that witnesses its own non-exhaustion. The mechanism is the same: self-action produces the structure that prevents completeness.

By the fundamental equivalence (Theorem 3.1), the converse also holds: every non-exhaustive system admits a π-preserving endomorphism (via the fiber swap construction). So the three classes—non-exhaustive, adequate, reflectively sufficient—coincide propositionally. The complete self-generation chain is:

$$\mathrm{ReflectivelySufficient} \xrightarrow{\text{twist produces witnesses}} \mathrm{Adequate}$$

$$\text{Adequate} \xrightarrow{\text{witnesses force non-exhaustion}} \text{NonExhaustive}$$

$$\text{NonExhaustive} \xrightarrow{\text{fiber swap}} \text{ReflectivelySufficient}.$$

The circle closes: there is no escape from any of the three without escaping all of them.

Remark 5.6 (Lean formalization).

- `AdequateReflectiveSystem`, `adequate_nonExhaustive`, `adequate_not_injective`, `identityCompare_not_adequate` in `Summit/Adequacy.lean`
- `ReflectivelySufficientSystem`, `toAdequate`, `reflectivelySufficient_nonExhaustive` in `Summit/SelfGeneration.lean`
- IC discharge: `icAdequateReflectiveSystem` in `Summit/ICAdequateDischarge.lean`

Remark 5.7 (Certification versus refinement). A potential confusion must be addressed. The resolution theory (§9) shows that the residual can always be eliminated by adding a sufficiently fine refinement. Does this contradict the claim that certification does not exhaust realization?

No. The distinction is between the *certification itself* (the fixed comparison map π) and *supplementary data* (a refinement $r : \mathsf{Realized} \to E$ added on top of π). For adequate systems, π is provably non-injective—the residual is intrinsic to the certification and cannot be eliminated by the certification itself (Proposition 5.3). A refinement can resolve the residual, but the refinement is extra structure *beyond* the certification. The identity refinement $r = \mathrm{id}$ always resolves, but it means "remember all of $\mathsf{Realized}$"—defeating the purpose of certification, which is to compress realization into something smaller.

The precise statement is the *certification-refinement gap theorem*: for any adequate system, (i) π is not injective, (ii) there exists a refinement that resolves, and (iii) no subsingleton-valued refinement suffices. The residual is real, its resolution requires genuine extra data, and the cost is measurable.

6 The residual kernel

Definition 6.1 (Residual kernel)**.** The *residual kernel* of an RCS S is

$$\mathcal{K}(\pi) \;=\; \big\{\, (x,y) \in \mathsf{Realized} \times \mathsf{Realized} \;\big|\; x \neq y \;\wedge\; \pi(x) = \pi(y) \,\big\}.$$

The kernel records every ordered pair of distinct realizations that certification cannot distinguish. It is a symmetric relation ($(x,y) \in \mathcal{K} \Leftrightarrow (y,x) \in \mathcal{K}$) and is intrinsic to π: two systems with the same comparison map have identical kernels.

From this point onward, non-exhaustion should no longer be thought of merely as the existence of two colliding realizations. Its canonical carrier is the residual kernel, and the rest of the theory consists in reading geometry, observability, and resolution from that single object.

Theorem 6.2 (Kernel characterization)**.**

(i) $\mathcal{K}(\pi) \neq \varnothing$ *if and only if* S *is non-exhaustive.*

(ii) $\mathcal{K}(\pi)$ *depends only on* π*.*

(iii) Two systems with the same π *have identical kernels, identical unresolved kernels for any refinement, and identical incompressibility at any type.*

(iv) $\mathcal{K}(\pi)$ *is the off-diagonal of the fiber product:* $\mathcal{K}(\pi) = (\mathsf{Realized} \times_{\mathsf{Bare}} \mathsf{Realized}) \setminus \Delta$*, where* $\mathsf{Realized} \times_{\mathsf{Bare}} \mathsf{Realized} = \{(x,y) \mid \pi(x) = \pi(y)\}$ *and* $\Delta = \{(x,x)\}$*. The system is exhaustive iff the fiber product equals the diagonal.*

Definition 6.3 (Veil)**.** For an adequate system A, the *veil* is the residual kernel $\mathcal{K}(\pi_A)$. By Theorem 5.2, the veil is nonempty. The adequate witnesses (w_1, w_2) are a canonical element of the veil.

Theorem 6.4 (Veil persistence)**.** *For any adequate system, the trivial refinement (to a point) leaves the veil fully intact: the unresolved kernel equals the full kernel.*

Remark 6.5 (Lean formalization).

- `ResidualKernel`, `residualKernel_nonempty_iff`, `residualKernel_symmetric`, `residualKernel_eq_of_compare_eq`, `residualTransfer_unresolvedKernel` in `Residual/ResidualKernel.lean`
- `adequate_kernel_nonempty`, `adequate_witnesses_in_kernel`, `adequate_veil_persists_trivial` in `Residual/ResidualKernel.lean`

7 Fiber geometry

The kernel decomposes fiberwise, and each fiber carries a maximally dense internal structure.

Definition 7.1 (Per-fiber kernel)**.** For $b \in \mathsf{Bare}$, the *per-fiber kernel* is $\mathcal{K}_b = \{(x, y) \mid x \neq y,\ \pi(x) = b,\ \pi(y) = b\}$. This is the off-diagonal of $\mathrm{Fib}(b)$.

Theorem 7.2 (Fiberwise decomposition)**.** $\mathcal{K}(\pi) = \bigcup_{b \in \mathsf{Bare}} \mathcal{K}_b$.

Theorem 7.3 (Complete-fiber theorem)**.** *For every $b \in \mathsf{Bare}$, the per-fiber kernel $\mathcal{K}_b$ is a* complete graph*: every pair of distinct elements in* $\mathrm{Fib}(b)$ *is a kernel edge.*

Proof. If $x, y \in \mathrm{Fib}(b)$ with $x \neq y$, then $\pi(x) = b = \pi(y)$, so $(x, y) \in \mathcal{K}_b$. □

Fibers are not merely clusters of collisions—they are *maximally dense indistinguishability cliques*. Every possible collision within a fiber actually occurs. There are no gaps in the residual within a fiber.

Theorem 7.4 (Connectivity theorem)**.** *Define $x \sim y$ iff $\pi(x) = \pi(y)$ (fiber equivalence). Then:*
(i) $\sim$ is an equivalence relation.
(ii) $x \sim y$ iff $x = y$ or $(x, y) \in \mathcal{K}(\pi)$.
(iii) The equivalence classes of $\sim$ are exactly the fibers $\mathrm{Fib}(b)$*.*
(iv) Fibers are the connected components of the kernel graph.

Proof. Part (i) is immediate. For (ii), $x \sim y$ means $\pi(x) = \pi(y)$; if $x \neq y$ this is a kernel pair, otherwise $x = y$. Part (iii) follows from the definition of fibers. Part (iv) follows from (ii)–(iii): two elements are reachable via kernel edges iff they share a fiber. □

The kernel neighbors of any element x are exactly $\mathrm{Fib}(\pi(x)) \setminus \{x\}$—all other members of its fiber.

Remark 7.5 (Lean formalization).
- `KernelAt`, `residualKernel_eq_iUnion_kernelAt`, `kernelAt_complete`, `kernelAt_eq_offDiagonal_fiber` in `Residual/KernelStratification.lean` and `Residual/MinimalResolution.lean`
- `KernelRel`, `KernelRelRefl`, `kernelRelRefl_equivalence`, `fiberEquiv_iff_eq_or_kernelRel`, `fiber_eq_kernelRelRefl_class`, `kernelNeighbors_eq_fiber_minus_self` in `Residual/KernelGraph.lean` and `Residual/MinimalResolution.lean`

8 Observable duality

The kernel determines a sharp boundary between what certification can and cannot see.

Definition 8.1 (Bare-determined predicate)**.** A predicate $P : \mathsf{Realized} \to \mathrm{Prop}$ is *bare-determined* if it factors through π: there exists $Q : \mathsf{Bare} \to \mathrm{Prop}$ with $P(x) \Leftrightarrow Q(\pi(x))$ for all x.

Theorem 8.2 (Observable algebra)**.** *The bare-determined predicates form a Boolean subalgebra of the full predicate algebra on* $\mathsf{Realized}$*: they are closed under $\wedge$, $\vee$, $\neg$, $\to$, $\leftrightarrow$, and contain $\top$ and $\bot$.*

Proof. If P factors through Q_P and P' factors through $Q_{P'}$, then $P \wedge P'$ factors through $Q_P \wedge Q_{P'}$, and similarly for the other connectives. □

Theorem 8.3 (Universal factorization)**.** *A function $f : \mathsf{Realized} \to \alpha$ (with α nonempty) is constant on every fiber of π if and only if it factors through π: there exists $g : \mathsf{Bare} \to \alpha$ with $f = g \circ \pi$.*

Proof. The forward direction constructs g classically: for each $b \in \mathsf{Bare}$, if b is in the image of π, set $g(b) = f(x)$ for any x with $\pi(x) = b$ (well-defined by the fiber-constancy hypothesis); otherwise, set $g(b)$ to an arbitrary element of α. The converse is immediate. □

This theorem says π is the *universal coarse observable*: every observable that cannot see inside fibers must factor through π, and anything coarser than π is blind to all kernel pairs.

Theorem 8.4 (Kernel-observable duality)**.** *A predicate $P : \mathsf{Realized} \to \mathrm{Prop}$ is not bare-determined if and only if it separates at least one kernel pair: there exist $(x, y) \in \mathcal{K}(\pi)$ with $(P(x) \wedge \neg P(y))$ or $(\neg P(x) \wedge P(y))$.*

Proof. Forward: if P is not bare-determined, it is not constant on some fiber (otherwise it would factor through π by Theorem 8.3 applied to the characteristic function). So there exist x, y in the same fiber with $P(x)$ and $\neg P(y)$; since they share a fiber and are distinguished by P, they are distinct, hence a kernel pair.

Backward: if P separates $(x, y) \in \mathcal{K}$, then P is not constant on $\mathrm{Fib}(\pi(x))$, so it cannot factor through π. □

This theorem completely characterizes the boundary between map-visible and map-invisible predicates. The residual observables are not an indeterminate remainder; they are exactly the predicates that cut across kernel pairs. The kernel and the non-bare observables are dual objects: each determines the other.

Remark 8.5 (Lean formalization).
- `BareDetermined`, `bareDetermined_and/or/not/imp`, `bareDetermined_of_comp` in `Residual/ObservableAlgebra.lean`
- `constant_on_fibers_iff_factors`, `factors_of_constant_on_fibers` in the same file
- `not_bareDetermined_iff_separates_kernel` in `Residual/ResidualKernel.lean`

9 Refinement and resolution

A *refinement* $r : \mathsf{Realized} \to E$ provides extra information beyond π. The *refined comparison* is the pair $(\pi, r) : \mathsf{Realized} \to \mathsf{Bare} \times E$.

Definition 9.1 (Resolution)**.** A refinement r *resolves* the system if the refined comparison (π, r) is injective.

Theorem 9.2 (Resolution criterion)**.** *A refinement resolves the system if and only if it separates every kernel pair: for all $(x, y) \in \mathcal{K}(\pi)$, $r(x) \neq r(y)$.*

Proof. If (π, r) is injective and $(x, y) \in \mathcal{K}$, then $\pi(x) = \pi(y)$, so injectivity forces $r(x) \neq r(y)$ (else $(\pi(x), r(x)) = (\pi(y), r(y))$ would give $x = y$, contradicting $x \neq y$). Conversely, if r separates every kernel pair and $(\pi(x), r(x)) = (\pi(y), r(y))$, then either $x = y$ (done) or $(x, y) \in \mathcal{K}$ and $r(x) = r(y)$, contradicting separation. □

Theorem 9.3 (Fiber-local resolution)**.** *Resolution decomposes into independent per-fiber problems: a refinement resolves the whole system if and only if it resolves every fiber individually. Cross-fiber pairs are never kernel pairs.*

Theorem 9.4 (Coloring interpretation)**.** *A refinement acts as a graph coloring of each fiber clique: it assigns a color $r(x) \in E$ to each vertex. It resolves a fiber if and only if it properly colors the complete graph on that fiber.*

Since each fiber kernel is a complete graph K_n on $n = |\mathrm{Fib}(b)|$ vertices, the chromatic number is n.

Theorem 9.5 (Resolution complexity)**.** *Resolving a fiber of size n requires at least n distinct refinement values on that fiber. Equivalently, the refinement must be injective on the fiber. This is the chromatic number of the complete graph K_n: the exact cost of eliminating the residual at a given fiber is its size.*

Definition 9.6 (Refined fiber class)**.** Given a refinement r, the *refined fiber class* at (b, e) is $\{x \in \mathsf{Realized} \mid \pi(x) = b \wedge r(x) = e\}$. The fiber $\mathrm{Fib}(b)$ is the disjoint union of its refined classes.

Theorem 9.7 (Terminal refinement)**.** *A refinement is terminal (fully resolving) if and only if every refined class is a singleton.*

Definition 9.8 (Refinement preorder)**.** Refinement r_2 *refines* r_1 if equal r_2-values imply equal r_1-values. This is a preorder with the trivial refinement (to a point) as bottom and the identity refinement (to $\mathsf{Realized}$) as top.

Definition 9.9 (Unresolved kernel)**.** The *unresolved kernel* of a refinement r is $\mathcal{K}_r^{\mathrm{unres}} = \{(x, y) \in \mathcal{K}(\pi) \mid r(x) = r(y)\}$.

Theorem 9.10 (Monotonicity and progress)**.**

(i) The unresolved kernel is antitone: finer refinements have weakly smaller unresolved kernels.
(ii) The resolved kernel (complement in $\mathcal{K}$) is monotone: finer refinements resolve weakly more pairs.
(iii) A refinement resolves iff its unresolved kernel is empty.
(iv) Any refinement separating at least one new kernel pair strictly shrinks the unresolved kernel.

Theorem 9.11 (Trivial-resolves-iff-exhaustive)**.** *The trivial refinement resolves if and only if the system is exhaustive. The cheapest possible resolution works exactly when there is no residual.*

Theorem 9.12 (Adequate resolution bound)**.** *For adequate systems: subsingleton-valued refinements cannot resolve. The identity refinement always resolves. So the resolution complexity of an adequate system is bounded below by 2 and above by $|\mathsf{Realized}|$.*

Definition 9.13 (Nontrivial locus)**.** The *nontrivial locus* of an RCS is the set of bare certificates whose fibers contain at least two distinct elements. The system is exhaustive iff the nontrivial locus is empty; non-exhaustive iff it is nonempty. The nontrivial locus is nonempty iff the kernel is nonempty.

The nontrivial locus is the "support" of the residual: it tells you *which* certificates carry hidden structure. The resolution cost at each certificate in the locus is the fiber size at that certificate.

Definition 9.14 (Persistent observable)**.** A non-bare-determined predicate is *persistent* under a refinement r if it separates a kernel pair that r also fails to separate (i.e., a pair in the unresolved kernel of r).

Theorem 9.15 (Persistence dichotomy)**.** *Either a refinement resolves (and no observable is persistent) or the unresolved kernel is nonempty (and persistent observables exist).*

Persistent observables are the predicates that remain "invisible to certification" even after the refinement is applied. They measure what the refinement has not yet resolved.

Theorem 9.16 (Certification-refinement gap)**.** *For any adequate system A:*

(i) π_A is not injective (the certification has a residual).
(ii) There exists a refinement that resolves (the identity refinement).
(iii) No subsingleton-valued refinement resolves (the resolution cost is ≥ 2).

The residual is intrinsic to the certification level: it is determined by π alone and cannot be changed by altering other system data. Eliminating it requires genuine extra data beyond the certification.

Remark 9.17 (Lean formalization).
- Key definitions: `Refinement`, `refinedCompare`, `RefinesTo`, `UnresolvedKernel`, `RefinedFiberClass`
- Key theorems: `resolves_iff_separates_all_kernel_pairs`, `resolves_iff_properly_colors_all`, `terminal_iff_all_singletons`, `trivial_resolves_iff_exhaustive`
- Spread across `Residual/ResidualStructure.lean`, `Residual/MinimalResolution.lean`, `Residual/ResolutionComplexity.lean`, `Residual/QuantitativeResidual.lean`, `Residual/RefinementOrder.lean`, `Residual/Incompressibility.lean`, `Residual/KernelStratification.lean`
- Fiber spectrum: `FiberTrivial`, `FiberNontrivial`, `NontrivialLocus` in `Residual/FiberSpectrum.lean`
- Categorical kernel: `FiberProduct`, `Diagonal`, `residualKernel_eq_fiberProduct_minus_diagonal` in `Residual/CategoricalKernel.lean`
- See `MANIFEST.md` for the complete index

10 Optimal certification and residual measurement

The resolution theory raises a natural question: if refinement can always eliminate the residual, does anything genuinely remain? The answer depends on what "remains" means.

Theorem 10.1 (Optimal refinement existence)**.** *For any RCS S, there exists a refinement that fully resolves the system: the identity refinement $r = \mathrm{id} : \mathsf{Realized} \to \mathsf{Realized}$ makes the refined comparison (π, id) injective.*

Proof. If $(\pi(x), x) = (\pi(y), y)$, then $x = y$. □

So the residual can always be eliminated. But the identity refinement means "remember all of Realized"—it defeats the purpose of certification, which is to compress realization into something smaller. The real question is the *cost* of resolution.

Theorem 10.2 (Zero residual characterization)**.** *For any RCS S, the following are equivalent:*
(i) $\mathcal{K}(\pi) = \varnothing$ (kernel empty).
(ii) π is injective.
(iii) Every fiber is trivial (at most one element).
(iv) The nontrivial locus is empty.
(v) The trivial refinement (to a point) resolves.
(vi) Every predicate on Realized *is bare-determined.*
When any of these hold, the residual is completely absent: certification sees everything, no extra data is needed, and there is no hidden structure.

Definition 10.3 (Refinement gap)**.** The *refinement gap* of a refinement r is its unresolved kernel: $\mathrm{Gap}(r) = \{(x, y) \in \mathcal{K}(\pi) \mid r(x) = r(y)\}$. This measures how far r is from full resolution.

Theorem 10.4 (Gap properties)**.**
(i) The gap is empty iff the refinement resolves.
(ii) The gap of the trivial refinement is the full kernel.
(iii) The gap of the identity refinement is empty.
(iv) Finer refinements have smaller gaps (monotonicity).

The gap gives a precise "distance to resolution": it is the set of kernel pairs that the refinement has not yet separated. Each step of refinement that separates a new kernel pair strictly shrinks the gap (Theorem 9.10(iv)).

Theorem 10.5 (Adequate irreducible residual)**.** *For any adequate system A:*

(i) The gap of the trivial refinement is nonempty (it equals the full kernel, which is nonempty by adequacy).

(ii) The gap of any subsingleton-valued refinement is nonempty.

(iii) The adequate witnesses (w_1, w_2) are in the gap of every subsingleton-valued refinement.

The residual of an adequate system has irreducible depth: it cannot be eliminated without genuine extra data.

Theorem 10.6 (Nothing remains at optimal)**.** *If π is injective, then:*

(i) $\mathcal{K}(\pi) = \varnothing$.

(ii) The nontrivial locus is empty.

(iii) Every predicate on Realized *is bare-determined—certification sees everything.*

(iv) The trivial refinement resolves—no extra data is needed.

At optimal certification, the veil is gone. There is no hidden structure.

The picture is now complete. Certification determines a comparison map. If the map is injective, the residual is absent and certification exhausts realization. If it is not—as the adequacy theorem forces for systems with witness diversity—the residual is a nontrivial geometric object: the kernel, decomposed into complete fiber cliques, with exact resolution cost. The residual can be resolved by adding extra data, but the cost is real and measurable, and for adequate systems it is always positive.

Remark 10.7 (Lean formalization).

- `optimal_refinement_exists`, `zero_residual_iff_injective`, `zero_residual_iff_all_trivial`, `zero_residual_iff_trivial_resolves` in `Residual/OptimalCertification.lean`
- `refinementGap`, `gap_empty_iff_resolves`, `gap_trivial_eq_kernel`, `gap_identity_empty`, `gap_antitone` in the same file
- `adequate_irreducible_residual`, `nothing_remains_at_optimal`, `certification_refinement_gap` in the same file

11 Functoriality and invariance

The residual geometry is intrinsic to the comparison map and transforms coherently under maps between systems.

Definition 11.1 (Comparison-compatible map)**.** A *comparison-compatible map* from S_1 to S_2 (on the same carriers) is a function $\varphi : \mathsf{Realized} \to \mathsf{Realized}$ with $\pi_2 \circ \varphi = \pi_1$.

Theorem 11.2 (Functoriality)**.**

(i) An injective compatible map sends kernel pairs to kernel pairs and preserves fiber membership.

(ii) Compatible maps compose: if $\varphi : S_1 \to S_2$ and $\psi : S_2 \to S_3$, then $\psi \circ \varphi : S_1 \to S_3$.

(iii) The identity is compatible with itself.

(iv) Composition is associative, and the identity is a two-sided unit.

(v) Two systems with the same π have identical kernels, identical unresolved kernels, and identical incompressibility.

Thus RCS on fixed carriers and compatible maps form a category.

Theorem 11.3 (Optimal coarsening)**.** *π is the finest fiber-constant map: any function constant on fibers factors through π. Any map coarser than π is blind to all kernel pairs.*

Theorem 11.4 (Witness independence)**.** *Fiber nontriviality, the fiber partition, and the bare-determined algebra all depend only on π, not on the choice of witnesses, canonical region, or soundness predicate.*

Remark 11.5 (Lean formalization).

- `CompareCompatible`, `compCompatible`, `idCompatible`, `compCompatible_assoc`, `compatible_preserves_kernel`, `CoarserThanCompare`, `compare_is_finest_fiber_constant`, `coarser_blind_to_kernel` in `Residual/FundamentalTheorem.lean`
- `nonExhaustive_depends_only_on_compare`, `fiber_eq_of_compare_eq`, `bareDetermined_depends_only_on_compare` in `Residual/WitnessFiber.lean`

12 The fundamental theorem of residual geometry

We package the entire residual theory into a single statement.

Theorem 12.1 (Residual Geometry Theorem)**.** *For any comparison map $\pi : \mathsf{Realized} \to \mathsf{Bare}$, the results of §§6–9 hold unconditionally: the kernel characterization (Theorem 6.2), fiberwise decomposition (Theorem 7.2), complete-fiber structure (Theorem 7.3), kernel-observable duality (Theorem 8.4), resolution criterion (Theorem 9.2), fiber-local resolution (Theorem 9.3), resolution complexity (Theorem 9.5), and the exhaustion criterion (Theorem 4.1). No hypotheses on the RCS are required.*

The adequacy and self-generation theorems (§5) tell us *when* the kernel is nontrivial; the fundamental theorem tells us *what* the kernel is and how it behaves, for every system.

Remark 12.2 (Lean formalization).
- `FundamentalResidualPackage`, `fundamentalResidualPackage` in `Residual/FundamentalTheorem.lean`
- The package is a Lean structure whose fields are the eight component theorems; the constructor `fundamentalResidualPackage` builds it for any RCS

13 Source discharges

Historically, the universal theory did not arise by beginning with arbitrary maps and asking abstract questions about fibers. It was forced by concrete theorems in three source programs. Infinity Compression exhibited irreducible residue above canonical certification; NEMS supplied a computability-facing non-erasure barrier; APS isolated a local-to-global realization architecture that prevented the synthesis from collapsing into a vacuous slogan. The present paper extracts from those discharges the universal map-theoretic core that survives after all domain-specific structure is stripped away.

Historically, the theory arose from these source programs; formally, they now serve as concrete discharges of a universal map-theoretic framework. The abstract theory developed in §§3–12 is universal: it applies to any comparison map. But the theory acquires force when discharged against concrete mathematical architectures that exhibit non-exhaustion for principled reasons. The program originated in a synthesis of three independent research lines, each governing one layer of a reflexive architecture.

The three-layer origin. The Strata program unifies three lines into a single architecture:

Layer	Source program	Structural role
Outer	NEMS	Admissibility of determinacy under closure
Middle	APS	Exact mechanics of uniform realization
Inner	Infinity Compression	Reflective residue above canonical certification

The *Non-Erasure Principle* proved in the Strata formalization states that under explicit cross-layer coherence hypotheses, outer semantic non-exhaustion and inner enriched irreducibility hold simultaneously and resist joint flattening. The *Unified Non-Erasure Law* gives the biconditional: enriched irreducibility $\Leftrightarrow$ existence of a semantic remainder, under coherence and

universal totality. These results are formalized in `Bridge/NonErasurePrinciple.lean` and `Bridge/UniversalNonErasureLaw.lean`.

Infinity Compression (IC). The IC flagship [16, 18, 17] provides the internal origin of the program and the primary discharge into the universal theory. Two distinct role-separated skeletons with the same summit component extraction yield the witness-diversity data for an adequate system. The IC adequate system forgets to the same RCS as the IC universal instance (`icAdequateReflectiveSystem_toRCS` = `rfl`). The role permutation that swaps the two skeletons is a non-identity π-preserving endomorphism, so IC also discharges into the reflectively sufficient class.

NEMS / computability. The No External Model Selection (NEMS) framework [11, 12, 14] provides the outer (computability) layer. The diagonal barrier and semantic remainder from NEMS instantiate the reflexive layer interface. NEMS record-truth undecidability serves as the concrete semantic remainder.

APS / algebra. The abstract indexed programming systems (APS) framework [15] provides the middle (realization) layer. The corrected exactness biconditional (indexed composition $\Leftrightarrow$ finite tracking + gluing) instantiates the realization layer interface.

Cross-corpus synthesis. The three source programs combine into a concrete linked architecture (`LinkedArchitecture`) with both coherence axioms proved definitionally. The cross-corpus biconditional (IC enriched irreducibility $\Leftrightarrow$ NEMS semantic remainder) and the universal non-erasure law hold for all spine-eligible pairs. This synthesis is the concrete evidence that the abstract residual geometry is not vacuous: it is instantiated by a real cross-domain architecture with machine-checked proofs at every level.

Remark 13.1 (Lean formalization).
- `icAdequateReflectiveSystem` in `Summit/ICAdequateDischarge.lean`
- `nemsReflexiveLayer` in `Instances/FromNEMS.lean`
- `apsRealizationLayerFromIff` in `Instances/FromAPS.lean`
- `crossCorpusLinkedArchitecture` in `Instances/CrossCorpusInstance.lean`
- `universal_nonerasure_law` in `Bridge/UniversalNonErasureLaw.lean`

14 Relationship to incompleteness and proof relevance

Relationship to Gödel's incompleteness. The fundamental equivalence has a structural parallel with Gödel's incompleteness theorems [1]: in both cases, a system's own internal resources produce the object that prevents its completeness. In Gödel's case, the system's expressive power constructs an undecidable sentence; in ours, the system's comparison geometry constructs a non-trivial fiber automorphism. The all-or-nothing dichotomy (Theorem 3.2) mirrors Gödel's result in that there is no partial escape: either the system is completely exhaustive, or it has all five forms of non-exhaustion simultaneously.

The differences are equally important. First, Gödel's theorem concerns provability of sentences; ours concerns distinguishability of realizations. The objects are different in kind. Second, Gödel's theorem is purely negative; ours has a large positive component—the residual has a canonical geometry with exact laws (the Fundamental Theorem of Residual Geometry, Theorem 12.1). Third, Gödel's theorem requires arithmetic expressiveness; the fundamental theorem of residual geometry is unconditional. The adequacy conditions that force nontriviality (§5) are separate from the geometric theory itself. The analogy is structural, not identificatory: the present theory is not a new incompleteness theorem about provability, but a theorem about the geometry of indistinguishability under maps.

The meta-level distinction. A natural question arises: if the residual can be measured exactly (§10), does this contradict Gödel's claim that a system cannot know its own incompleteness?

No. The residual kernel is defined in terms of *both* Bare and Realized—it requires access to the realization level, which is exactly what certification compresses away. Any predicate that factors through π (a "bare-level" predicate) is provably constant on every kernel pair: it cannot detect whether the residual exists. Formally: for any $Q : \mathsf{Bare} \to \mathrm{Prop}$ and any kernel pair (x, y), $Q(\pi(x)) = Q(\pi(y))$.

So the residual is *invisible from the bare level* but *visible from the meta-level* (which has access to both Bare and Realized). This is precisely the Gödelian structure: the veil exists and has exact geometry, but the certification level cannot see it. An external observer—or a stronger system with access to the realization level—can measure the residual exactly. The certification itself cannot.

The theory does not let a system measure its own incompleteness. It lets a *meta-theory* measure the incompleteness of any system it can see from outside. That is the correct reading of the resolution and measurement results in §10.

Related mathematical structures. The fiber decomposition of a map is classical; the residual kernel is the complement of the diagonal in the fiber product $\mathsf{Realized} \times_{\mathsf{Bare}} \mathsf{Realized}$ [4]. The observable algebra is a quotient Boolean algebra in the sense of Stone duality [2]. The coloring interpretation connects to the classical theory of chromatic numbers of complete graphs [3]. The functoriality results are elementary category theory [4]. What is new is not any single construction but the *synthesis*: the complete package of kernel, observables, refinement, coloring, and complexity as a unified theory of information loss under arbitrary maps, with the Classification Theorem and all-or-nothing dichotomy as the governing structural results.

Why this is not dependent-type proof relevance. In dependent type theory [6, 7], "proof relevance" means that different proofs of the same proposition can be distinguished. Our non-exhaustion is superficially similar: different realizations of the same bare certificate are distinguished. But the resemblance is shallow. Proof relevance is a design choice in a type system; our non-exhaustion is a *theorem* about comparison maps, with no dependence on any particular foundational framework. The residual kernel, observable duality, and resolution theory have no analogues in the proof relevance literature. The theory is about the geometry of what *any* certification process forgets, not about the internal structure of proofs.

15 The universal theory of forgetting

The results of this paper are stated for an arbitrary comparison map $\pi : \mathsf{Realized} \to \mathsf{Bare}$ between arbitrary types. Nothing in the definitions or theorems requires that π arise from a "certification process." The word "certification" is a motivating interpretation, not a formal hypothesis.

This means the entire theory applies to *any function between any two types.* Every such function determines a residual kernel, a fiber decomposition, an observable algebra, a resolution theory, and exact complexity bounds. The fundamental equivalence, the all-or-nothing dichotomy, and the meta-level invisibility theorem hold universally.

Theorem 15.1 (Universal forgetting)**.** *For any function $\pi : E \to B$ between types in a common universe:*

(i) The residual kernel $\{(x, y) \mid x \neq y \wedge \pi(x) = \pi(y)\}$ is the canonical object of what π forgets.

(ii) Non-injectivity, witness diversity, fiber automorphism, and nontrivial kernel are equivalent (fundamental equivalence).

(iii) A predicate on E factors through π iff it does not separate any kernel pair (observable duality).

(iv) A refinement $r : E \to C$ makes (π, r) injective iff r separates every kernel pair (resolution criterion).

(v) Resolving a fiber of size n requires r to be injective on that fiber (resolution complexity).

(vi) Any predicate $Q : B \to \mathrm{Prop}$ satisfies $Q(\pi(x)) = Q(\pi(y))$ whenever $\pi(x) = \pi(y)$: the codomain level is blind to the residual (meta-level invisibility).

(vii) The full fundamental residual package and five-way equivalence exist unconditionally.

This theorem applies simultaneously to:

Domain	π	Residual
Group theory	homomorphism $\varphi : G \to H$	classical kernel $\ker \varphi$
Topology	quotient $X \to X/\sim$	equivalence classes
Type theory	projection $\Sigma_{x:A} B(x) \to A$	fiber family B
Physics	measurement state $\to$ observable	unmeasured degrees of freedom
Compilation	compiler source $\to$ bytecode	erased source structure
Cryptography	hash data $\to$ hash	collision structure

In each case, the residual kernel is the set of distinct inputs that the function identifies. The observable duality says which predicates can and cannot see the lost information. The resolution complexity says how much extra data is needed to recover it. And the meta-level theorem says the lost information is invisible from the codomain.

The theory is therefore not merely a theory of certification failure. It is a *universal theory of information loss under any map*—a geometry of forgetting that applies wherever a function compresses structure.

Remark 15.2 (Scope and non-claims). We do not claim that every useful map is best studied via this theory, nor that all information-loss phenomena are exhausted by the kernel. We do not claim automatic recoverability of the residual from the codomain: the meta-level theorem (§14) proves the opposite. We do not claim that the theory replaces domain-specific tools (e.g., cohomological obstructions in group extensions, or Rice-style barriers in computability). The claim is that the residual kernel, observable duality, and resolution theory provide a *universal structural layer* that applies to any map, complementing rather than replacing domain-specific analysis (e.g., [10] for group cohomology, [5] for computability barriers).

Remark 15.3 (Lean formalization).

- `rcsOfMap`, `forgettingKernel`, `universal_fundamental_equivalence`, `universal_observable_duality` in `Residual/UniversalForgetting.lean`
- `universal_resolution`, `universal_meta_blindness`, `every_function_has_residual_geometry`, `every_function_has_five_way_equivalence` in the same file

16 Frontier

The theory as presented is complete at the qualitative and structural level. We distinguish immediate mathematical next steps from deeper programmatic conjectures.

Immediate next steps

Residual spectra and moduli. The fiber-size spectrum (the multiset of fiber sizes) is a complete invariant of resolution complexity. Defining entropy-like measures of residual spread, moduli of kernel structure, and spectral invariants that survive bounded refinement would turn the theory from structural to quantitative.

Residual observables under refinement. Which non-bare-determined predicates survive a given refinement? Can one classify the "persistent residual observables" that remain non-bare-determined after any bounded refinement? This is the formal version of asking what structure is *robustly* invisible to certification.

Iterated certification. What happens when a refined comparison is itself treated as a new comparison map and the theory is applied again? Does the residual stabilize, shrink monotonically, or exhibit more complex dynamics?

Category-theoretic packaging. Compatible maps compose associatively with identity units (Theorem 11.2). The next step is to package RCS and compatible maps into a category and ask whether the kernel construction defines a functor, and whether the fundamental residual package is natural.

Deeper conjectures

Domain-specific kernel structure. In each discharge domain (group extensions, embedding problems, Quillen fibers, computability barriers), the kernel has domain-specific structure beyond the abstract theory. Rewriting these domain results in kernel language would test the theory's power as a unifying instrument.

Broader inevitability. The current summit classes require explicit structural data. Whether there exist broader "richness" conditions that force witness diversity or fiber automorphisms from more abstract principles—without supplying witnesses or twists as data—remains the deepest open question.

17 Conclusion

This paper has established that information loss under any map is not a deficiency to be lamented but a geometric phenomenon to be studied. The residual kernel is the canonical object of that geometry; the fundamental equivalence theorem shows that non-injectivity, witness diversity, fiber automorphism, and nontrivial kernel are the same phenomenon viewed from five angles; and the fundamental theorem of residual geometry gives the complete structure of the residual unconditionally for every function between any two types.

The theory opens a new kind of mathematical question. Classical algebra asks: given a group, what are its subgroups? Classical topology asks: given a space, what are its open sets? The residual theory asks: given *any function*, what is the exact geometry of what it forgets? The answer—a disjoint union of complete indistinguishability cliques, with observables dual to kernel separation and resolution governed by graph coloring—is both precise and universal.

The formalization in Lean 4 with zero `sorry` ensures that every claim in this paper is machine-checked. The discharge against the Infinity Compression flagship, the NEMS computability anchor, and the APS algebraic layer ensures that the abstract theory is not vacuous.

What a non-injective map forgets is not an accident, not an informal remainder, and not merely a by-product of examples. It is a canonically organized object with exact laws. This paper identifies that object, proves its classification, determines its observables, and computes the exact cost of resolving it. The problem is no longer whether information is lost under a map—it always is, when the map is non-injective. The problem is to understand the geometry of that loss. This paper gives that geometry.

A Theorem index and Lean correspondence

The table below lists every named paper result, its Lean theorem name, and its module. All names are within the `ReflexiveArchitecture.Universal` namespace; `Summit/*` live under `ReflexiveArchitecture.Universal.Summit` and `Residual/*` under `ReflexiveArchitecture.Universal.Residual`.

Paper result	Lean name	Module
Classification Theorem (Thm. 3.1)	`equiv_1_iff_2` ... `equiv_1_iff_5`	`Summit/FundamentalEquivalence`
All-or-nothing dichotomy (Thm. 3.2)	`all_or_nothing`	`Summit/FundamentalEquivalence`
Five-way equivalence package	`FiveWayEquivalence`, `fiveWayEquivalence`	`Summit/FundamentalEquivalence`
Fiber automorphism diagonal	`nonExhaustive_implies_fiberAutomorphism`	`Summit/FundamentalEquivalence`
Vanishing Criterion (Thm. 4.1)	`exhaustive_iff_kernel_empty`, `kernel_empty_iff_all_fibers_subsingleton`	`Residual/FundamentalTheorem`
Adequacy Theorem (Thm. 5.2)	`adequate_nonExhaustive`	`Summit/Adequacy`
Self-generation (Thm. 5.5)	`reflectivelySufficient_nonExhaustive`, `toAdequate`	`Summit/SelfGeneration`
Kernel characterization (Thm. 6.2)	`residualKernel_nonempty_iff`, `residualKernel_eq_of_compare_eq`	`Residual/ResidualKernel`
Fiberwise decomposition (Thm. 7.2)	`residualKernel_eq_iUnion_kernelAt`	`Residual/KernelStratification`
Complete-fiber theorem (Thm. 7.3)	`kernelAt_complete`	`Residual/MinimalResolution`
Connectivity theorem (Thm. 7.4)	`kernelRelRefl_equivalence`, `fiber_eq_kernelRelRefl_class`	`Residual/KernelGraph`
Observable algebra (Thm. 8.2)	`bareDetermined_and`, `bareDetermined_or`, `bareDetermined_not`, ...	`Residual/ObservableAlgebra`
Universal factorization (Thm. 8.3)	`constant_on_fibers_iff_factors`	`Residual/ObservableAlgebra`
Kernel-observable duality (Thm. 8.4)	`not_bareDetermined_iff_separates_kernel`	`Residual/ResidualKernel`
Resolution criterion (Thm. 9.2)	`resolves_iff_separates_all_kernel_pairs`	`Residual/ResidualKernel`
Resolution complexity (Thm. 9.5)	`resolution_complexity_theorem`	`Residual/FundamentalTheorem`
Certification–Refinement Gap (Thm. 9.16)	`certification_refinement_gap`	`Residual/OptimalCertification`
Residual Geometry Theorem (Thm. 12.1)	`FundamentalResidualPackage`, `fundamentalResidualPackage`	`Residual/FundamentalTheorem`
Persistence dichotomy (Thm. 9.15)	`persistence_dichotomy`, `exists_persistent_of_unresolved`	`Residual/MetaLevel`
Universal forgetting (Thm. 15.1)	`every_function_has_residual_geometry`, `every_function_has_five_way_equivalence`	`Residual/UniversalForgetting`
Nothing remains at optimal (Thm. 10.6)	`nothing_remains_at_optimal`	`Residual/OptimalCertification`
Category laws (Thm. 11.2)	`RCSCategoryLaws`, `rcsCategoryLaws`	`Residual/RCSCategory`

Build: `lake build` from `reflexive-architecture-lean/`. See `MANIFEST.md` for the complete listing (∼235 theorems, 26 modules).

References

[1] K. Gödel, *Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I*, Monatshefte für Mathematik und Physik, 38:173–198, 1931. English translation:

On Formally Undecidable Propositions of Principia Mathematica and Related Systems, Basic Books, 1962.

[2] M. H. Stone, *The theory of representations for Boolean algebras*, Transactions of the American Mathematical Society, 40(1):37–111, 1936.

[3] R. L. Brooks, *On colouring the nodes of a network*, Mathematical Proceedings of the Cambridge Philosophical Society, 37(2):194–197, 1941; C. Berge, *Graphs and Hypergraphs*, North-Holland, Amsterdam, 1973.

[4] S. Mac Lane, *Categories for the Working Mathematician*, Springer-Verlag, New York, 1971.

[5] H. G. Rice, *Classes of recursively enumerable sets and their decision problems*, Transactions of the American Mathematical Society, 74(2):358–366, 1953.

[6] P. Martin-Löf, *Intuitionistic Type Theory*, Bibliopolis, Naples, 1984.

[7] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013. Available at `https://homotopytypetheory.org/book/`.

[8] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in *Automated Deduction — CADE 28*, Lecture Notes in Computer Science vol. 12699, Springer, 2021, pp. 625–635.

[9] The Mathlib Community, *The Lean Mathematical Library*, in *Proceedings of CPP 2020*, ACM, 2020, pp. 367–381.

[10] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics vol. 87, Springer-Verlag, New York, 1982.

[11] N. Spivack, *The NEMS Framework: No External Model Selection*, Paper 1, NEMS Suite, 2026. Lean: `nems-lean`; zero `sorry`, zero custom axioms.

[12] N. Spivack, *From NEMS to MFRR: A Machine-Checked Bridge Between Semantic Closure and Reflexive Reality*, Paper 8, NEMS Suite, 2026. Lean: `nems-lean`; zero `sorry`, zero custom axioms (8051 jobs).

[13] N. Spivack, *No Emulation, Self-Necessitating Adjudication*, Paper 15, NEMS Suite, 2026. Lean: `nems-lean`; zero `sorry`, zero custom axioms (8069 jobs).

[14] N. Spivack, *Big Results from the NEMS Program*, Paper 81, NEMS Suite (Synthesis), 2026. Lean: `nems-lean`.

[15] N. Spivack, *Composition as Finite Tracking plus Gluing, and its Separation from Recursion in Abstract Indexed Programming Systems*, 2026. Lean: `aps-recursion-uniformization-lean`; zero `sorry`, zero additional axioms.

[16] N. Spivack, *Canonical Certification Does Not Exhaust Reflective Structure*, Infinity Compression Suite, 2026. Lean: `infinity-compression-lean`; zero `sorry`.

[17] N. Spivack, *Certification, Realization, and Obstruction—A Universal Fiber Architecture*, Infinity Compression Suite (Synthesis Map), 2026. Lean: `infinity-compression-lean`.

[18] N. Spivack, *Reflective Non-Exhaustion Summit*, Infinity Compression Suite, 2026. Lean: `infinity-compression-lean`.

[19] N. Spivack, *Closure, Realization, and Reflective Residue*, Reflexive Architecture / Strata, 2026. Lean: `reflexive-architecture-lean`; zero `sorry`, zero custom axioms (∼8270 jobs).